

Dear Hiring Team at the National Research Council Canada,

I am writing to express my strong enthusiasm for the AI Research Student position within the Digital Technologies Research Center at the National Research Council (NRC) for the Winter 2027 term in Ottawa. As a Software Engineering Co-op student at McGill University with hands-on experience designing deep learning models from scratch, building computer vision pipelines, and developing custom visualization interfaces, I am eager to contribute to your research in drone swarm visualization and LLM/VLM explainability.

My technical and research background aligns directly with the core objectives of this project. I have built custom neural networks in Python in the food industry using standard PyTorch, NumPy, and scikit-learn libraries, and I am currently engineering an end-to-end computer vision and NLP system to perform real-time image recognition and multi-modal data synthesis. Furthermore, during my research work at the National Institute for Scientific Research (INRS), I engineered automated Python algorithms to extract spatial intelligence and analyze complex multi-angle datasets. This research foundation equips me with the technical rigor needed to develop multi-view visualization tools for drone swarms and integrate interpretability layers for Vision-Language Models (VLMs) and Large Language Models (LLMs).

In addition to machine learning development, I possess full-stack programming capabilities (Python, JavaScript/React, C++) that enable me to build responsive visual dashboards and diagnostic tools for decision support systems. Combined with my formal background in project management and public sector research experience, I bring a systematic approach to conducting literary reviews, benchmarking model interpretability, and presenting technical research outcomes.

The NRC's mission to advance Canadian technological innovation deeply resonates with me. I am excited about the opportunity to apply my machine learning, computer vision, and visualization experience to support your R&D initiatives in drone swarm intelligence and explainable AI.

Thank you for your time and consideration. I welcome the opportunity to discuss how my AI/ML research background aligns with your team's objectives.

Sincerely,

Renji Jenkins